Quantum Mechanics-1 (Assignment-3)

(Dated: 19th November, 2024)

1. Let $\vec{v} = (v_1, v_2, v_3)$ be a three-dimensional unit vector and $\vec{\sigma} = (\sigma_1, \sigma_2, \sigma_3)$, where $\sigma_1, \sigma_2, \sigma_3$ are Pauli spin matrices. Define $\vec{v} \cdot \vec{\sigma} \equiv v_1\sigma_1 + v_2\sigma_2 + v_3\sigma_3$.

   (a) Show that $\vec{v} \cdot \vec{\sigma}$ has eigenvectors $|\vec{v}; \pm\rangle = \cos\frac{\theta}{2}|0\rangle + e^{i\phi}\sin\frac{\theta}{2}|1\rangle$ corresponding to ± 1 eigenvalues. Here, $|0\rangle$ and $|1\rangle$ are eigenvectors of σ_3 with $+1$ and -1 eigenvalues respectively. And, θ, ϕ are polar and azimuthal angles respectively.

   (b) Show that the projectors onto the eigenspace corresponding to these eigenvalues are $P_\pm = \frac{I \pm \vec{v}\cdot\vec{\sigma}}{2}$.

   (c) Let $f : \mathbb{C} \to \mathbb{C}$ be a function and let θ be a real number. Show that

$$f(\theta\,\vec{v}\cdot\vec{\sigma}) = \frac{f(\theta) + f(-\theta)}{2} I + \frac{f(\theta) - f(-\theta)}{2}\vec{v}\cdot\vec{\sigma}$$

2. **a.** The simplest way to derive the Schwarz inequality goes as follows. First, observe

$$(\langle\alpha| + \lambda^* \langle\beta|)(|\alpha\rangle + \lambda\,|\beta\rangle) \geq 0$$

   for any complex number λ; then choose λ in such a way that the preceding inequality reduces to the Schwarz inequality.

   b. Show that the equality sign in the generalized uncertainty relation holds if the state in question satisfies

$$\Delta A|\alpha\rangle = \lambda \Delta B|\alpha\rangle$$

   with λ purely imaginary.

   c. Explicit calculations using the usual rules of wave mechanics show that the wave function for a Gaussian wave packet given by

$$\langle x'|\alpha\rangle = (2\pi d^2)^{-1/4} \exp\left[\frac{i\langle p\rangle x'}{\hbar} - \frac{(x' - \langle x\rangle)^2}{4d^2}\right]$$

   satisfies the minimum uncertainty relation

$$\sqrt{\langle(\Delta x)^2\rangle}\,\sqrt{\langle(\Delta p)^2\rangle} = \frac{\hbar}{2}.$$

   Prove that the requirement

$$\langle x'|\Delta x|\alpha\rangle = (\text{imaginary number})\langle x'|\Delta p|\alpha\rangle$$

   is indeed satisfied for such a Gaussian wave packet, in agreement with (b).

3. The translation operator for a finite (spatial) displacement is given by

$$\hat{T}(\mathbf{l}) = \exp\left(\frac{-i\mathbf{p}.\mathbf{l}}{\hbar}\right),$$

   where $\mathbf{p}$ is the momentum operator.

   (a) Evaluate

$$[x_i, \hat{T}(\mathbf{l})].$$

   (b) Using (a) demonstrate how the expectation value $\langle\mathbf{x}\rangle$ changes under translation.

4. Two particles have angular momentum $j_1 = 1$ and $j_2 = 1$. Write down the possible basis vectors $\{|j_1, j_2, j, m\rangle\}$ which are common eigenvectors of J_1, J_2, J^2, J_z. Express them in terms another set of basis vectors $\{|j_1, j_2, m_1, m_2\rangle\}$ which are common set of eigenvectors for J_1, J_2, J_{1z}, J_{2z}.

5. Two particles have angular momentum $j_1 = 1$ and $j_2 = 1/2$.

(a) Calculate the CG coefficients corresponding to the four vectors of total angular momentum $j = 3/2$ block only.

(b) We know that,

$$d^{(j)}_{mm'}(\beta) = \sum_{m_1 m_2} \sum_{m'_1 m'_2} \langle j_1, j_2; m_1, m_2 | j, m \rangle \langle j_1, j_2; m'_1, m'_2 | j, m' \rangle \, d^{(j_1)}_{m_1 m'_1}(\beta) \, d^{(j_2)}_{m_2 m'_2}(\beta),$$

where $d^{j}_{mm'}(\beta) = \langle jm' | \exp\{-iJ_y\beta/2\} \, |jm\rangle$. Now, from the knowledge of $d^{1/2}(\beta)$ and $d^{1}(\beta)$ from class, find out the matrix $d^{3/2}(\beta)$.